



74.13

$$\left. \begin{aligned} V_2 - V_1 &= V_T \ln \frac{I_2}{I_1} \\ I_1 &= 1 \text{ mA}, V_1 = 0.7 \text{ V} \\ I_2 &= ? \quad V_2 = 0.8 \text{ V} \end{aligned} \right\} \Rightarrow I_2 = I_1 \cdot e^{\frac{100 \text{ mV}}{25 \text{ mV}}} = 64.5 \text{ mA}$$

$$\therefore R = \frac{10 - 2.4 \text{ V}}{64.5 \text{ mA}} = 139 \Omega$$

74.19

a) Diode 电流变化 1 mA 时 V_D 变化 5 mV $\Rightarrow r_d = 5 \Omega$

$$r_d = \frac{V_T}{I_D} \quad \left\{ \Rightarrow I_D = \frac{25 \text{ mV}}{5 \Omega} = 5 \text{ mA} \right.$$

$$\therefore R = \frac{15 - 3}{5 \text{ mA}} = 2.4 \text{ k}\Omega$$

b) $I_D = I_S e^{\frac{V_D}{V_T}}$

$$\left. \begin{aligned} I_D &= 5 \text{ mA} \\ V_T &= 25 \text{ mV} \\ V_D &= \frac{3}{4} \text{ V} \end{aligned} \right\} \Rightarrow I_S = 4.679 \times 10^{-16} \text{ A}$$

c) $I_L = 1 \text{ mA}$ 时, 使流过 Diode 的电流为 $5 - 1 = 4 \text{ mA}$

$$\text{此时对应的 } V_D = V_T \ln \frac{I_D}{I_S} = 744.421 \text{ mV}$$

$$\Delta V_o = 744.421 \times 4 - 3000 = -22.316 \text{ mV}$$